

NF EN 206+A2 / DREUX-GORISSE

قدهتسملا تاغصاوملاو قيسدنهللا تارشؤملا

FCK,28 (CHARACTERISTIC)

25 MPa

Target fcm: 34.8 MPa

WATER / BINDER (W/C)

0.52

Dreux G: 0.45

TARGET SLUMP

7 cm

Class S2

FRESH DENSITY

2435 kg/m³

Air: 1.5%

PROJECT & SITE CONTEXT

Project Name: New_Concrete_Project
Client / Owner: Client Organization
Batching Plant: Central Batching Plant
Location / Site: Central Construction Site

CONCRETE CLASS & ENVIRONMENT

Standard Class: C25/31 (EN 206)
Exposure Class: XC1
Control Class: NORMAL
Placement Method: Crane / Bucket

FORMULATION PARAMETERS

Design Method: Georges Dreux-Gorissee
Max Aggregate (Dmax): 15 mm
Concrete Type: NSC
Cement CEM I 42.5 N يداي يذنانرو، تنمسا
Batch Calculation: 1 m³

CONSTITUENT MATERIALS & PHYSICAL PROPERTIES

TESTED SPECIFICATIONS

Component	Commercial Name / Specification	Source / Brand	Density	Absorption	Moisture	Notes / Limits
Cement (Liant)	SYS-CEM-001	GICA / Lafarge	3.10 g/cm ³	-	-	Class 42.5 MPa
Sand (Sable 0/4)	SYS-SND-002	Local Quarry	2.62 g/cm ³	1.4%	1.2%	FM: 2.55
Gravel (Gravier 4/20)	SYS-GRA-002	Regional Quarry	2.66 g/cm ³	1.1%	0.9%	Dmax: 15 mm
Mixing Water (Eau)	Potable Tap Water (قېمومعلا عيزونلا قكيش برش مايم)	Municipal / Well	1.00 g/cm ³	-	-	pH: 7.0
Admixture (Adjuvant)	Superplasticizer (يحملا قئاو ندلم)	Sika / MasterGlenium	1.06 g/cm ³	-	-	Dosage: 1.0%
Mineral Addition (SCM)	GGBFS (قباذ راجدلا بكرم) قموغلا يلاء طيشنم ديدج مثير	Industrial Mineral	2.92 g/cm ³	-	-	Sub: 20.0%
Fibers (Fibres)	Hooked-End Steel Fibers (فارطلا فوقعم يذلاوف قبايلا)	Specialty Fiber	7.85 g/cm ³	-	-	Dosage: -

THEORETICAL DRY PROPORTIONS (PER 1.0 M³ & BATCH OF 1 M³)

DREUX-GORISSE COMPOSITION

Constituent Material	Absolute Volume (L/m ³)	Dry Mass (kg/m ³)	Batch (1 m ³)	% Total Mass	Engineering Ratio
Cement (C)	96.8 L	300 kg	300.0 kg	12.4%	Binder Base (1.00 C)
Effective Water (E)	155.0 L	155.0 kg	155.0 kg	6.4%	W/C = 0.52
Dry Sand (Sable 0/4)	332.8 L	872 kg	872.0 kg	36.2%	G/(S+G) = 46.5% Sand
Dry Gravel (Gravier 4/20)	381.6 L	1015 kg	1015.0 kg	42.1%	Dmax = 15 mm
Slag	—	67 kg	67.0 kg	2.8%	Alternative binder
Superplasticizer (يحملا قئاو ندلم): Admixture	2.78 L	3.00 kg	3.00 kg	0.12%	0.8% of Binder
TOTAL DESIGN MASS (DRY / EFFECTIVE BASIS)	1000.0 L	2412 kg/m ³	2412.0 kg	100.0%	Compaction Gamma: 0.82

SITE MOISTURE CORRECTION & READY-MIX BATCHING SCALE WEIGHTS & AGGREGATE ADJUSTMENT

Material / Scale Point	Dry Mass (kg/m³)	Moisture (w%)	Moisture Delta (kg)	Actual Wet Scale (1 m³)	Batch Scale (1 m³)
Sand Scale Weight (Sable)	872 kg	1.2%	+10 kg	882 kg/m³	882.0 kg
Gravel Scale Weight (Gravier)	1015 kg	0.9%	+9 kg	1024 kg/m³	1024.0 kg
Mixer Water to Add (Eau ajoutée)	155 L	Apport Net	+4 L	159 L/m³	159.0 L
Cement Scale Weight (Ciment)	300 kg	0.0% (Sec)	0 kg	300 kg/m³	300.0 kg

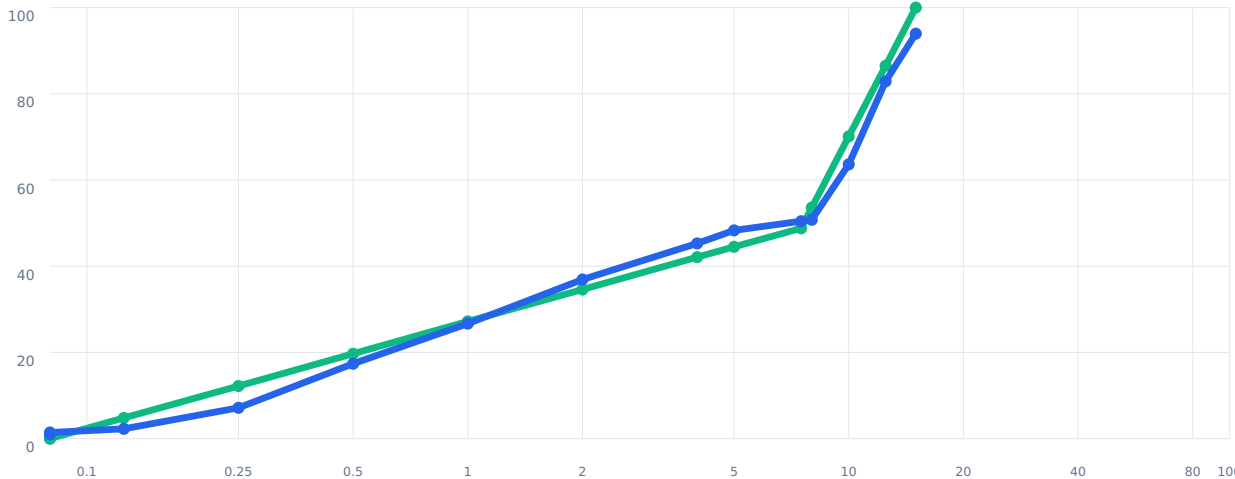
MECHANICAL PREDICTIONS & NORMATIVE COMPLIANCE MATRIX (EN 206 / ACI 211)QUALITY AUDIT

Engineering Parameter	Calculated / Measured	Design Requirement / Limit	Reference Standard	Status
Characteristic Compressive Strength (fck,28)	25 MPa	Target fcm: 34.8 MPa (Margin +9.8)	fcm >= fck + 1.64 sigma	CONFORMING
Water / Binder Ratio (E/C)	0.52	Limit: <= 0.50 (XC1)	NF EN 206 Table F.1	WARNING
Minimum Binder Content (kg/m³)	367 kg/m³	Limit: >= 300 kg/m³	NF EN 206 Table F.1	CONFORMING
Model Early Strength at 2 Days (fcm,2d)	15.7 MPa	Indicative only — do not use for stripping without test evidence	Hydration model / no lab result	MODEL PREDICTION
Predicted Characteristic Strength at 7 Days (fck,7d)	17.5 MPa	Characteristic curve; confirm by laboratory test	Model series / standard curing 20°C	MODEL PREDICTION
Flexural Tensile Strength (fctm)	2.56 MPa	fctm = 0.30 * fck^(2/3)	Eurocode 2 Eq. 3.1	THEORETICAL
Secant Modulus of Elasticity (Ecm)	32.0 GPa	Ecm = 22 * (fcm/10)^0.3	Eurocode 2 Table 3.1	THEORETICAL

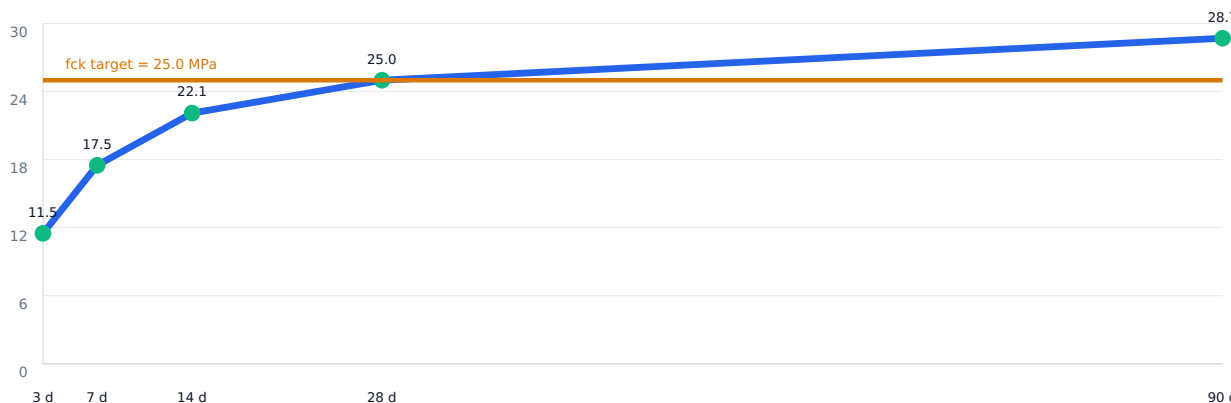
DREUX + MATERIAL BLEND

قيسدينهلا تاينحنملاو ي نايدلا ليلا

جريدتلا ي نحنم: Dreux قطلخلا ي لعللا جريدتلا ل باقم فدهتسملا
Target Dreux
Actual blend



رمعلا عم طغضلا ةمواقم روط



STRUCTURED SNAPSHOT

قعارملاو عبتلا

لجسلا ةلا

ي نحنملا

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ي فاه طاقءلا مة Snapshot/JSON قعارملا ل باق مظنم

ENGINEER / MATERIALS TECHNICIAN

Name: Senior Concrete Formulation Engineer
Date: 2026-10-08
Status: Prepared for technical review

[DIGITALLY VERIFIED]

LABORATORY DIRECTOR / QUALITY ASSURANCE

Authority: Director of Technical & Quality Control
Report control: Review + trial mix required
Cert Ref: SNO-MIX-CEM-I-42-5-N-25-20261008

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ENGINEERING REVIEW
TECHNICAL REVIEW